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Design of BLDCM servo controller based on TMS320F28377D

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Abstract. To meet the application requirements of high-precision and fast-response thrust vector control from a solid rocket motor, a digital and integrated servo controller for four brushless direct current motors was designed and implemented. The controller hardware system proposed in this paper includes control circuit module and drive circuit module. The former mainly includes power conversion circuit, DSP system circuit, A/D circuit, SPI communication interface circuit, and SCI communication interface circuit. The later mainly includes three-phase full-bridge six-arm drive circuit and optocoupler isolation circuit. The three closed-loop control strategies of position, speed and current based on PID control algorithm are adopted, and the mathematical model of the motor is established in Matlab/Simulink for verification. The structure composition and working principle of the servo controller are briefly introduced, and the hardware design scheme, software workflow and control strategy are described in detail. Experimental results show that the response speed and control accuracy of the proposed controller meet the technical requirements.

1. Introduction

Compared with the traditional DC motor, the brushless DC motor (BLDCM) adopts permanent magnets to replace the excitation winding coils in the rotor structure, and cancels the brush and the commutation device. With the characteristics of light weight, high efficiency and long lifetime, the BLDCM has been widely applied in aerospace, transportation, industry, agriculture and other fields [1-4].

For an aircraft with a solid rocket motor (SRM) as its power system, the thrust vector control technology is applied through adjusting the opening of servo valves driven by motors to obtain thrust vectors of different angles and sizes, which can effectively improve the maneuverability [5]. In order to achieve high-precision thrust vector control, the servo controller of a SRM needs to have fast response ability, high-precision position tracking ability and strong anti-interference ability, which puts forward higher requirements for the design scheme and working characteristics of the motor controller.

In this paper, an integrated and digital servo controller is proposed and implemented with the DSP TMS320F28377D as the main control chip. Displacement signals output by four displacement sensors are received through an externally expanded 4-channel and 16-bit ADC chip AD7606. The controller software compares actual position signals of four valves with the set target position signals, and then generates four position deviation signals. The PID control algorithm generates four PWM wave signals according to the obtained position deviation signals, and then drives BLDCMs move to their



target positions. At the same time, DSP collects the rotor position signal output by Hall angle sensor through its SPI module to realize electronic commutation without brush.

2. Controller Composition and Working Principle

2.1. Controller Scheme

The controller is composed of an upper PCB and a lower PCB in hardware structure [6]. The upper PCB is the control circuit module, and the lower PCB is the drive circuit module. The control circuit module mainly includes power conversion circuit, DSP system circuit, analog signal acquisition and conditioning circuit, SPI communication interface circuit, and SCI communication interface circuit. The drive circuit module mainly includes a three-phase full-bridge six-arm drive circuit and an optocoupler isolation circuit. The composition block diagram of the controller is shown in Figure 1.

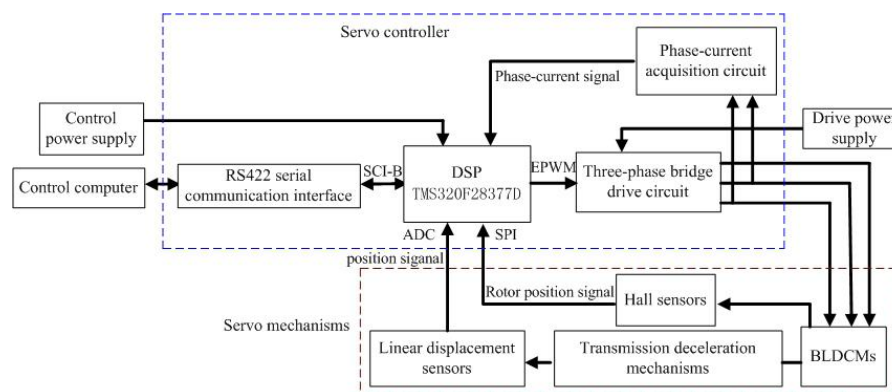


Figure 1. Composition block diagram of the controller.

The DSP system circuit receives control instructions given by a control computer, collects 1 bus voltage signal, 1 bus current signal, 8 phase current signals, 4 motor rotor position signals, 4 servo valve displacement signals, etc. and then outputs 24 PWM signals generated by control algorithm. The controller adopts high-performance TMS320C28X series 32-bit floating-point DSP processor TMS320F28377D as its main control unit. Information exchange is realized with the control computer through the isolated full-duplex RS422 serial interface [7].

2.2. BLDCM Mathematical Model

For a brushless DC motor with star-connected three-phase windings, ignoring factors such as cogging, eddy current loss and magnetic circuit saturation, its mathematical model can be obtained according to the working principle of the motor. Figure 2 is the BLDCM simulation model established in Matlab/Simulink, with which the working characteristics of the motor can be effectively analyzed.

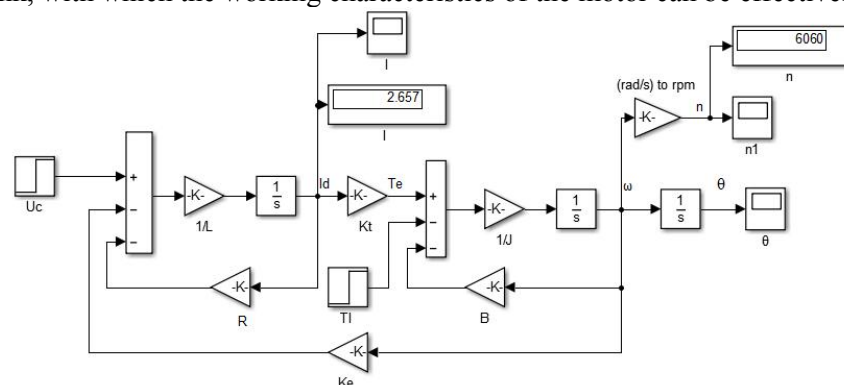


Figure 2. Simulink simulation model of brushless DC motor.

In Figure 2, U_c is the control voltage, L is the armature loop inductance, R is the armature loop resistance, K_t is the torque constant, K_e is the back EMF constant, B is the friction coefficient, J is the mass inertia, and T_l is the load torque. Under the action of U_c , the armature circuit I_d is generated, and then I_d generates the electromagnetic torque T_e . Under the combined action of electromagnetic torque, load torque and friction torque, the motor generates rotational speed ω , and ω is integrated to obtain the rotation angle θ . B is estimated according to equation (1), where I_0 and n_0 are the armature current and speed of the motor at no load, respectively.

$$B = 9.55 K_t I_0 / n_0 \quad (1)$$

3. Controller Hardware Designs

3.1. Control Unit

The control chip of the proposed controller selects the DSP TMS320F28377D, which is a dual-core architecture with two 32-bit CPUs, 1 MB of FLASH and 128 KB of RAM [8]. The chip has abundant peripheral resources, therefore it can simplify software development process, shorten development cycle, and reduce hardware costs.

The working principle diagram of the control unit is shown in Figure 3. CPU1 and CPU2 can output 12 PWM signals respectively, which can drive 2 motors respectively. CPU1 collects 4 displacement signals output by 4 linear displacement sensors via expanding an AD7606 chip. The displacement signals are calculated with the time information to obtain 4 velocity signals of the valve cores. CPU2 realizes the acquisition of 8 phase currents, 1 bus voltage and 1 bus current through the 12-bit ADC module on the DSP chip. CPU1 realizes the acquisition of 2 Hall angle signals through SPIC module and McBSP module, CPU2 realizes the acquisition of another two Hall angle signals through SPIA module and SPIB module. Hall angle sensors detect the motor angle data through coaxial connecting with the motor rotors.

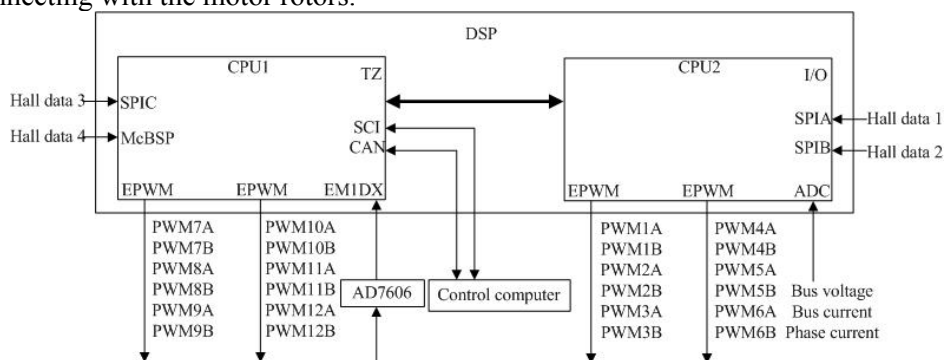


Figure 3. Working principle diagram of the control unit.

3.2. Drive Circuit

A three-phase full bridge drive circuit is designed with the integrated driver IR2136S from the IR company, which has high integration, few peripheral circuit components, simple structure, hence it improves the reliability of the circuit [9]. The schematic diagram of the three-phase bridge drive circuit is shown in Figure 4. The diodes V101, V102, and V103 are fast recovery diodes. And the capacitors C117, C118, and C119 are bootstrap capacitors, which together with the MOSFETs of the lower bridge arm realize the bootstrap function. The high-power switch tube in the drive circuit adopts Fairchild's high-power N-channel enhanced field effect transistor FQA65N20, with a rated overcurrent value of 65 A, a peak current of 260 A, and a drain-source rated voltage (VDSS) of 200 V, and its performance meets the requirements of the controller in this paper.

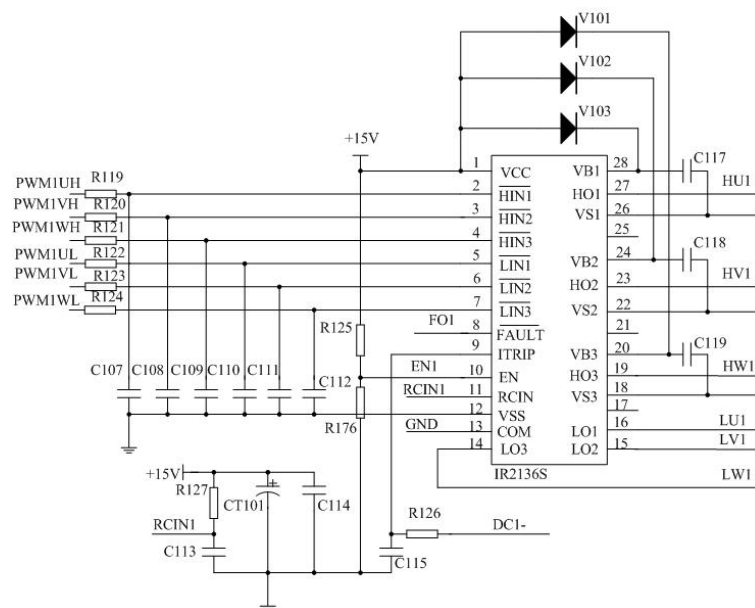


Figure 4. Schematic diagram of three-phase bridge drive circuit.

3.3. Current Acquisition Circuit

The A/D module with 20 channels and 12-bit precision on the DSP chip can realize high-speed synchronous sampling of bus voltage, bus current and phase current. The current sampling circuit adopts the Hall current sensor ACS710 with isolation function, which has the advantages of small size, high precision, good linearity, and single power supply. The ACS710 can be directly connected to the power output line of the three-phase bridge drive circuit to realize the acquisition of phase current. At the same time, in order to increase the signal-to-noise ratio in its output signal, a second-order low-pass filter is designed to regulate the output signal. The schematic diagram of the motor phase current sampling circuit is shown in Figure 5.

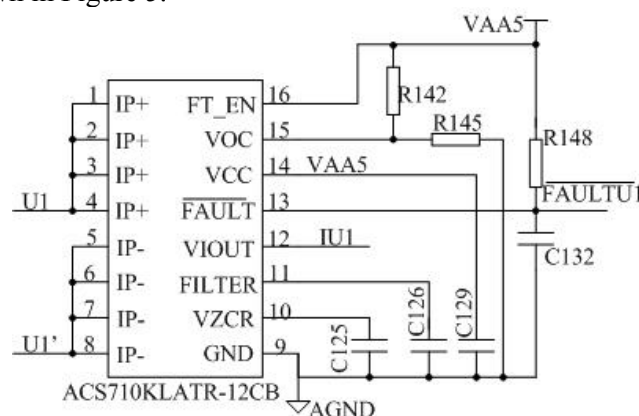


Figure 5. Schematic diagram of phase current acquisition circuit.

3.4. Displacement Signal Acquisition Circuit

The AD7606 chip of AD Company is a high-integration, 4-channel 16-bit precision synchronous sampling ADC. The voltage range of the output signal of the linear displacement sensor is $-5\text{ V} \sim 5\text{ V}$, and the voltage signal is input to the analog input pin of AD7606 through the emitter follower circuit. In the displacement signal acquisition circuit, set the sampling voltage of AD7606 to $\pm 5\text{ V}$. DSP collects the output signal of AD7606 through the parallel interface, in order to obtain higher sampling and transmission rate. The position signal sampling rate is set to 20 k/s .

3.5. Optocoupler Isolation Circuit

In order to reduce the influence of the chopping noise generated by the motor in the control circuit, an optocoupler isolation circuit is designed between the control circuit and the drive circuit. The PWM signal generated by the control circuit is input to the drive circuit after passing through the buffer circuit and the optocoupler isolation circuit. At the same time, the overcurrent signal, overheat signal and undervoltage signal output by the drive circuit are also isolated by the optocoupler isolation circuit. The high-speed optocoupler isolation chip selects HCPL0534, and the schematic diagram of the PWM signal optocoupler isolation circuit is shown in Figure 6.

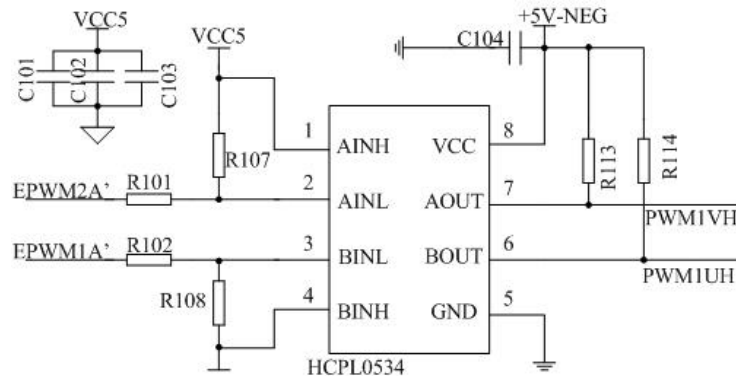


Figure 6. Schematic diagram of PWM optocoupler isolation circuit.

4. Controller Software Design

4.1. Software Function

Control software runs in the control circuit and is an embedded software without an operating system. The software adopts dual-core and dual-thread design scheme, which can drive four motors to rotate. CPU1 and CPU2 use a same design architecture, and can control two motors respectively. The main functions of controller software are to initialize CPU1 and CPU2, enable EPWM modules to output 24 PWM signals, configure the SCIB module to realize a RS422 communication port, and set three SPI modules to collect 3-channel of Hall sensor signals. At the same time, the software configures one McBSP module as a SPI module to collect the signal output by the fourth Hall sensor. One bus voltage signal, one bus current signal and 8 phase current signals are sampled by configuring the on-chip ADC. Four displacement signals are collected by expanding one AD7606 chip through the EMIF module, and the data exchange between CPU1 and CPU2 is realized through the internal information sharing area. The block diagram of software function is shown in Figure 7.

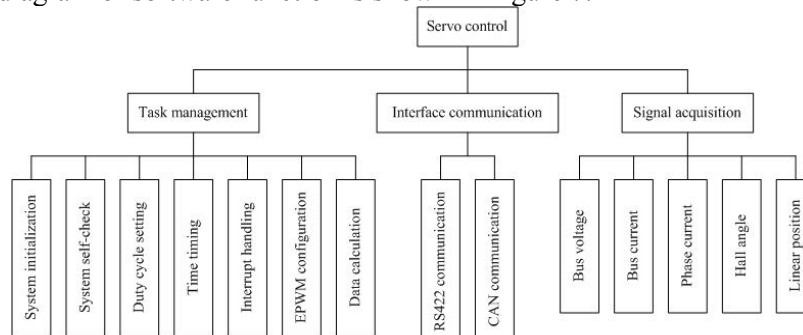


Figure 7. Block diagram of embedded software function.

Based on the CCS integrated development environment, C language is used to develop the controller software. The main flow chart of the software is shown in Figure 8. In main loop, the interrupt mechanism is applied to respond to emergencies, and various functions are realized by

calling subfunctions. Subroutines mainly include the initialization function of each functional module, the EPWM interrupt function, the signal acquisition function, the rotation speed calculation function, the serial port communication function, etc. In speed calculation interrupt function, the displacement signals collected by the AD7606 chip are calculated to obtain speed signals and a PID control algorithm is executed. In the main program, the speed data of the 4 motors is obtained by calling the speed calculation interrupt function. The sampling periods of the current loop, velocity loop and position loop are 50 μ s, 300 μ s and 400 μ s, respectively.

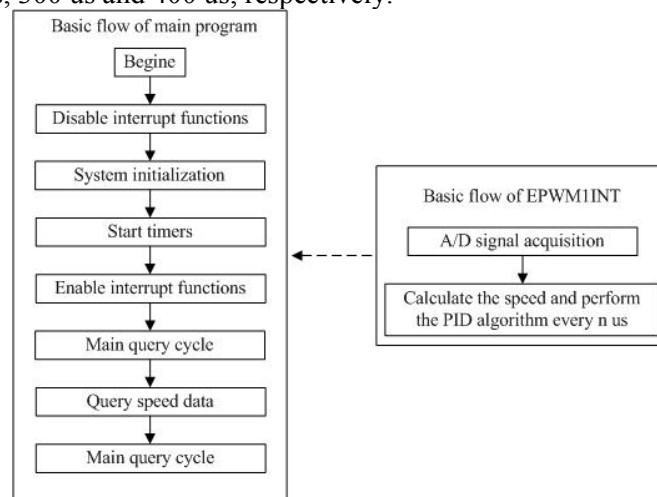


Figure 8. Main flow chart of control software.

4.2. Control Strategies

In order to obtain higher stability and control accuracy, the servo controller selects a three closed-loop control strategy when designing its control algorithm. The sequence from the outer loop to the inner loop is the position loop, the speed loop and the current loop. The block diagram of the three closed-loop control strategy is shown in Figure 9, which is mainly composed of a position controller, a speed controller, a current controller, a power inverter, a brushless DC motor, and modules such as current feedback, commutation control, speed calculation, and position feedback.

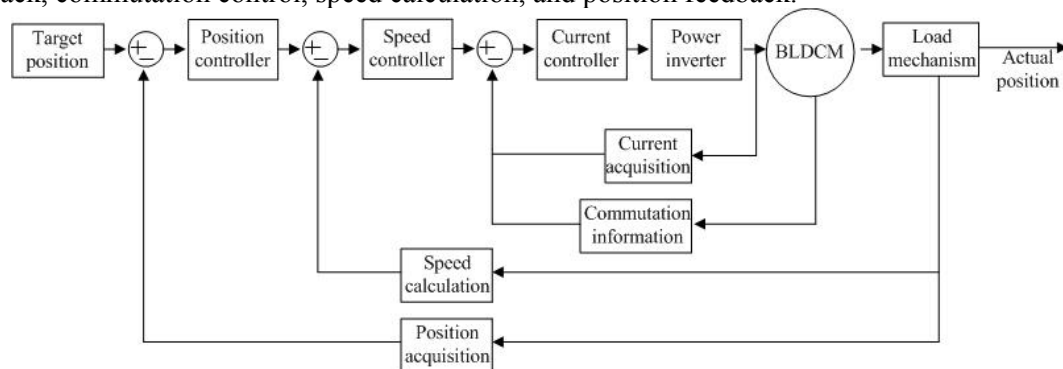


Figure 9. Principle diagram of the three closed-loop system.

The current loop is designed to maximize the dynamic performance of the motor. In order to improve the dynamic performance of the current loop response, the PI controller needs to be used to correct the system to a typical I-type system. The quality of the speed loop design directly affects the speed of the motor speed response. The function of the speed loop is to make the motor quickly reach the rated speed when the motor starts, and can quickly reduce the speed when the servo mechanism is about to reach the target position, so as to reduce the system overload. Therefore, the speed loop

adopts PI controller. To ensure the fast tracking of the system to the target position with small overshoot and steady-state error, the position loop control strategy adopts PID controller.

The PID controller parameters are designed and implemented by the PID tuning tool in the Simulink module separately from the inner loop to the outer loop. Then it is necessary to optimize the parameters of each controller according to the experimental results.

5. Experiment and results

5.1. Experiment System

The experimental system of the prototype is shown in Figure 10, which is mainly composed of control power supply, drive power supply, servo controller, computer, brushless DC motor and load mechanism, etc. When the controller was tested, the target position commands with different phases but the same amplitude and frequency were applied to the four servo valves respectively. The virtual instrument software for receiving and displaying test data was designed by LabVIEW, which is a graphical programming language of NI.

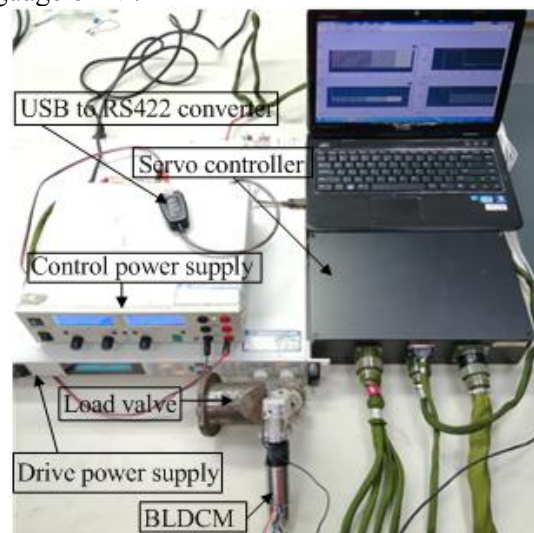


Figure 10. Controller prototype experimental system.

5.2. Results

The amplitude and frequency of the position control commands are 4 mm and 2.5 Hz, respectively. Figure 11 shows the position following test results of four servo valves, by calculation, the rise time is about 7.5 ms to 9 ms, and the dead time is about 2.5 ms to 3 ms, which show that the prototype has better control characteristics. Under no-load conditions, there is no position overshoot, but each servo valve has a certain degree of rebound effect, which will be improved during the SRM gas experiment.

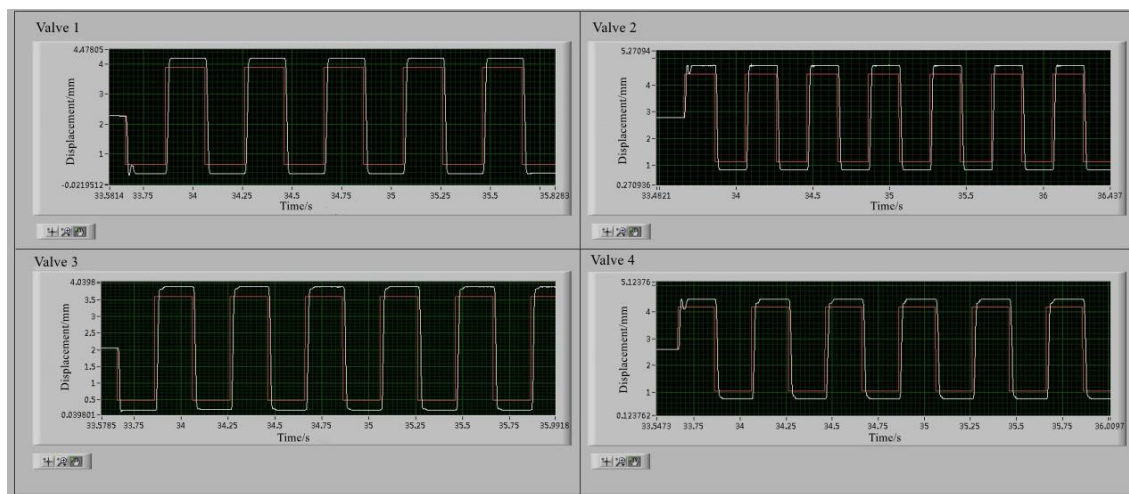


Figure 11. Test results of servo controller prototype.

6. Conclusions

In this paper, TMS320F28377D is selected as the main control unit, linear displacement sensor and current sensor are used as the main feedback detection units, and a solid rocket motor servo controller is designed adopting three closed-loop control strategies. By establishing a Matlab/Simulink model for the motor, the parameters of each closed-loop controller are determined, through which the control parameters of the prototype controller are designed. Experimental results show that the response speed and position control accuracy of the controller meet the technical requirements.

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